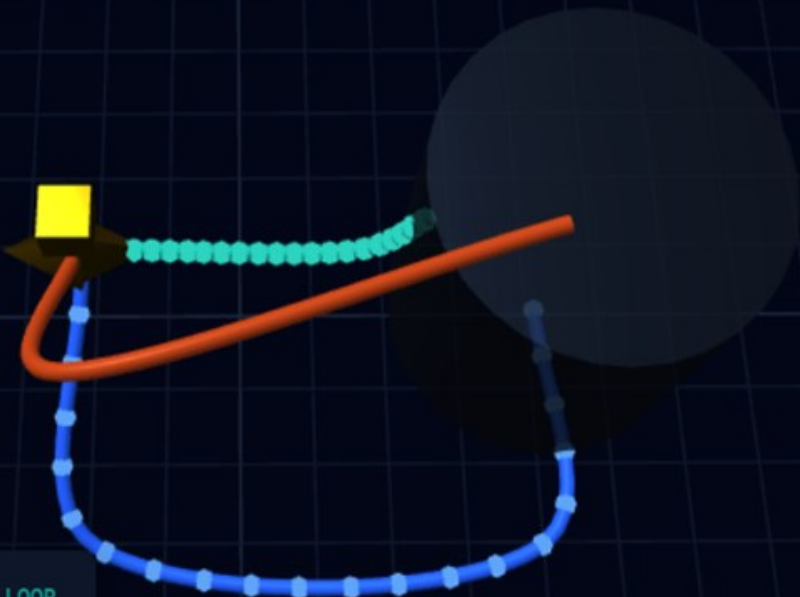


SYSTEM PIPING & INSTRUMENTATION SCHEMATIC (P&ID)

Scapush Precision Loop Modeling | Automated Three-Way Thermal Distribution

PROCESS SLIDE 4 INTERACTIVE



TAG: XV-101 (3-Way Valve) JACKET_LOOP
BIOLOGICAL TEMP STATE: 37.5°C NOMINAL
STEAM LINE FLUID V: 2.4 m/s
CONDENSATE RETURN: RECLAIM_ACTIVE

VALVE STATE SELECTION

ROUTE A: ENGINE JACKET THERMAL
HARVESTING

ACTIVE

ROUTE B: DIRECT STEAM INJECTION CORE

OFF

PROCESS DESCRIPTION

Engine Jacket Loop: High-efficiency thermal capturing channels heat continuously to the biological digester core to preserve structural 37.5°C stasis. Zero external parasitic fuel is required.

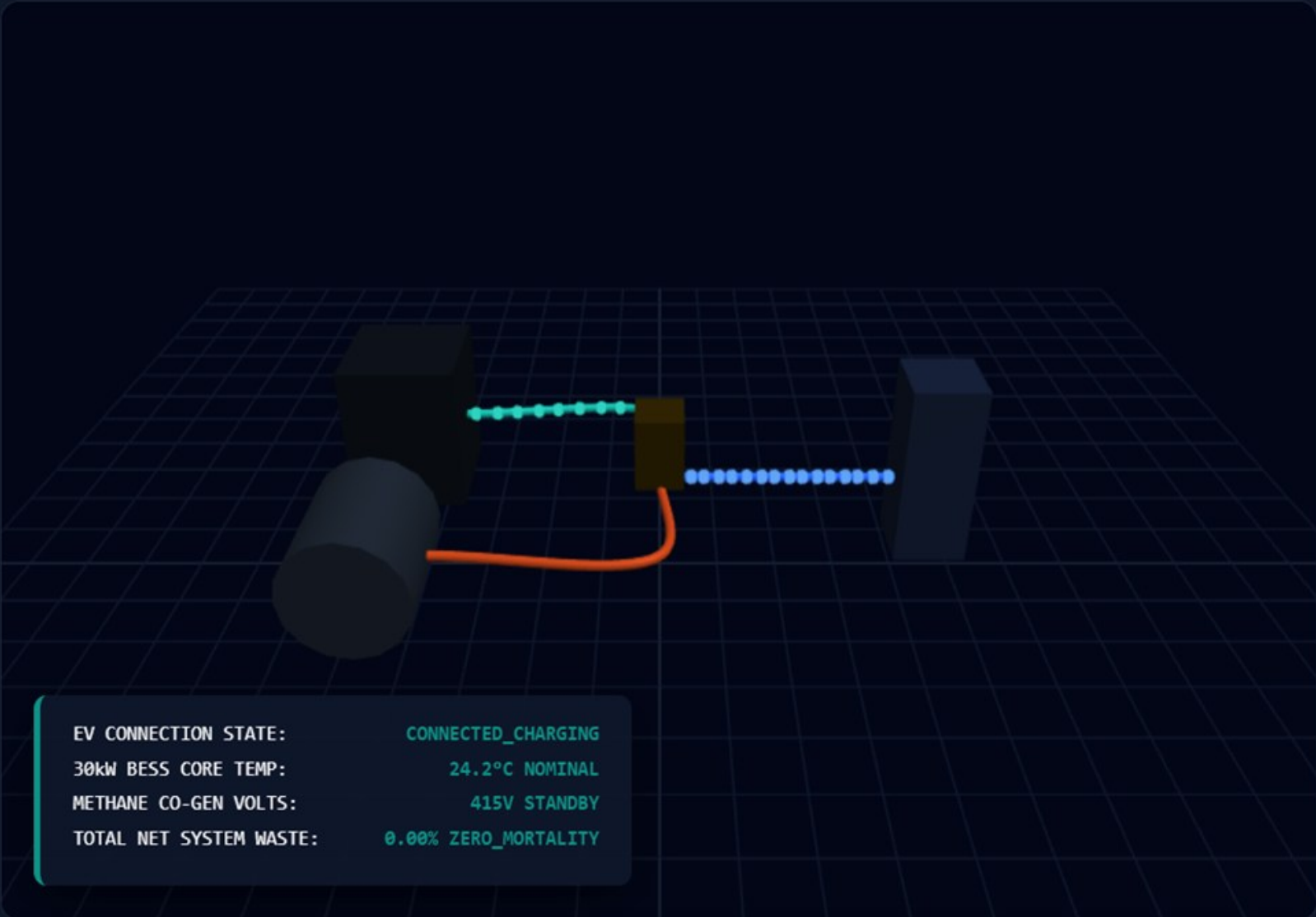
P&ID SYMBOLOGY LEGEND

- High-Temperature Thermal Loop Line
- High-Velocity Steam Injection Route
- Closed Condensation Recirculation Loop
- Automated Actuator Unit (XV-101)

BIO-SOLAR EV AUTOMATION INTERFACE

Lwandile Closed-Loop Blueprint | Zero-Waste Grid Synchronization

SCAPUSH PRECISION 2026



GRID SOURCE BALANCING

MODE A: PEAK DAY SOLAR + BESS FAST CHARGE

ACTIVE

MODE B: NIGHT-TIME BIO-THERMAL CO-GEN RUN

OFF

AUTOMATION MATRIX LOGIC

Peak Day Solar: The 30kW Solar BESS Core streams directly into the high-power DC charging nexus, providing Conny's EV point with zero-grid, carbon-shielded energy allocation.

GRID DISTRIBUTION LEGEND

- 30kW Solar & BESS Bus Line
- Methane Co-Generation Thermal Vector
- High-Voltage DC EV Charger Distribution Line
- Smart Micro-Grid Inverter PLC Controller

● PLC ACTIVE // STATION_NODE_01

NETWORK: LoRaWAN_ONLINE PID STATE: STABLE

TI-101 BIOREACTOR TEMP

37.48

°C

XV-101 VALVE POSITION

50.0

%



-5% STEP

+5% STEP

BIOGAS FLOW RATE

14.8

Nm³/h

SAFETY INTERLOCKS

[ESD_1: NORMAL]
[PRESSURE: ISOBARIC]
[OVERPRESSURE: SAFE]

REAL-TIME SYSTEM PROCESS CURVES

— Core Temp (°C) — Valve Position (%)

